SMART WASTE SORTING BIN

Standalone ESP32-S3-CAM · MobileNetV2 · sensor fusion · 4 streams

V2 / HK$1,000

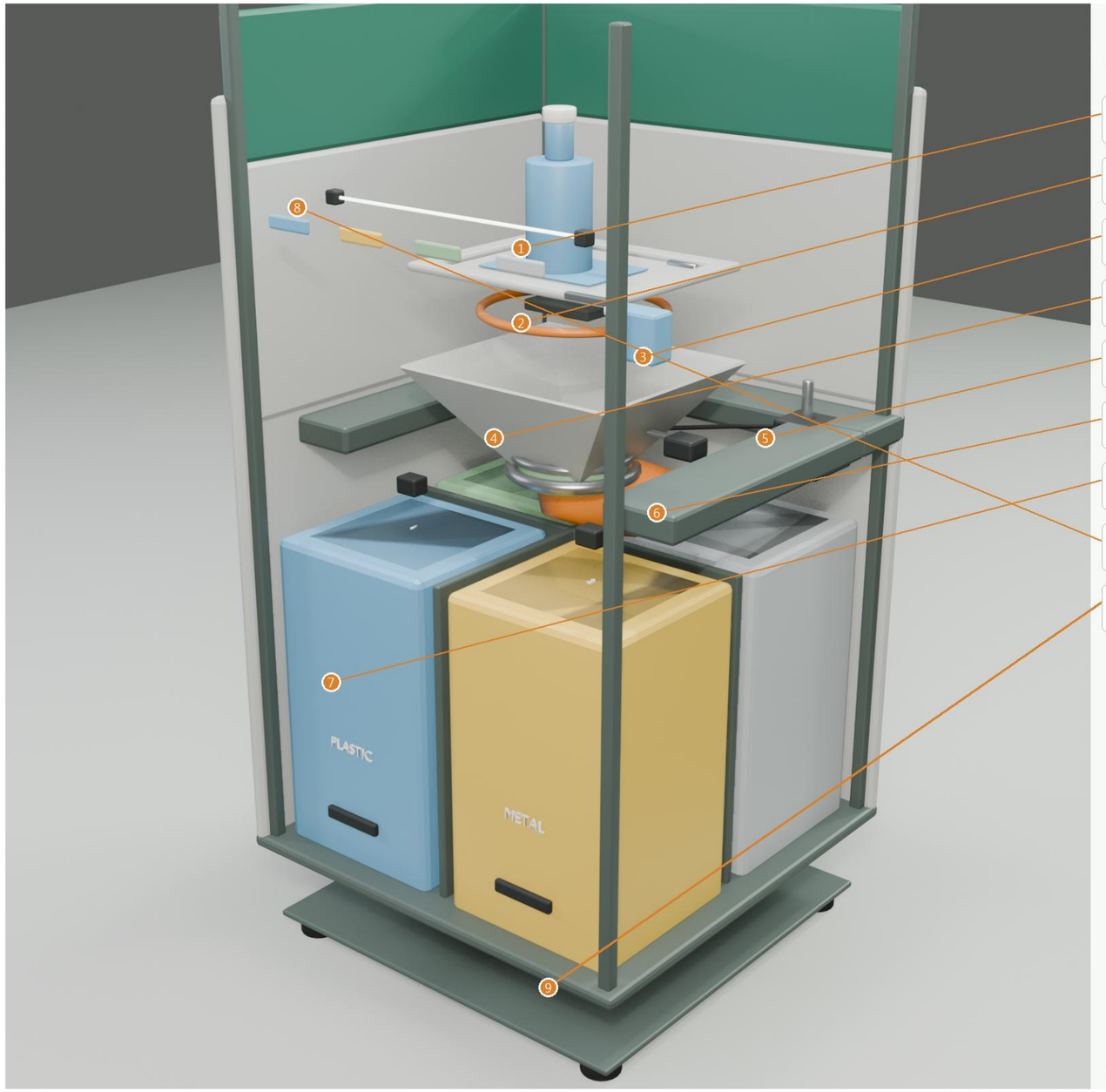


Design intent

Item	Value
Overall	420 × 420 × 900 mm
Bins	4 × 170 × 170 × 300 mm (8.7 L)
Total capacity	approx. 34.7 L
Platform	220 × 180 mm; 180 × 140 mm door
Funnel / spout	220 × 200 → 75 mm; 75 mm elbow
Max item	500 mL bottle / 330 mL can; <=0.5 kg
Cycle	6–10 s
BOM	HK$866
Total incl. 15%	approx. HK$996

No laptop / no cloud / no agent

ESP32-S3-CAM runs MobileNetV2 0.35/0.5 at 96–128 px, INT8.
Sensor fusion: camera + 5 kg load cell + moisture + IR.
WiFi AP and OLED are for status only.



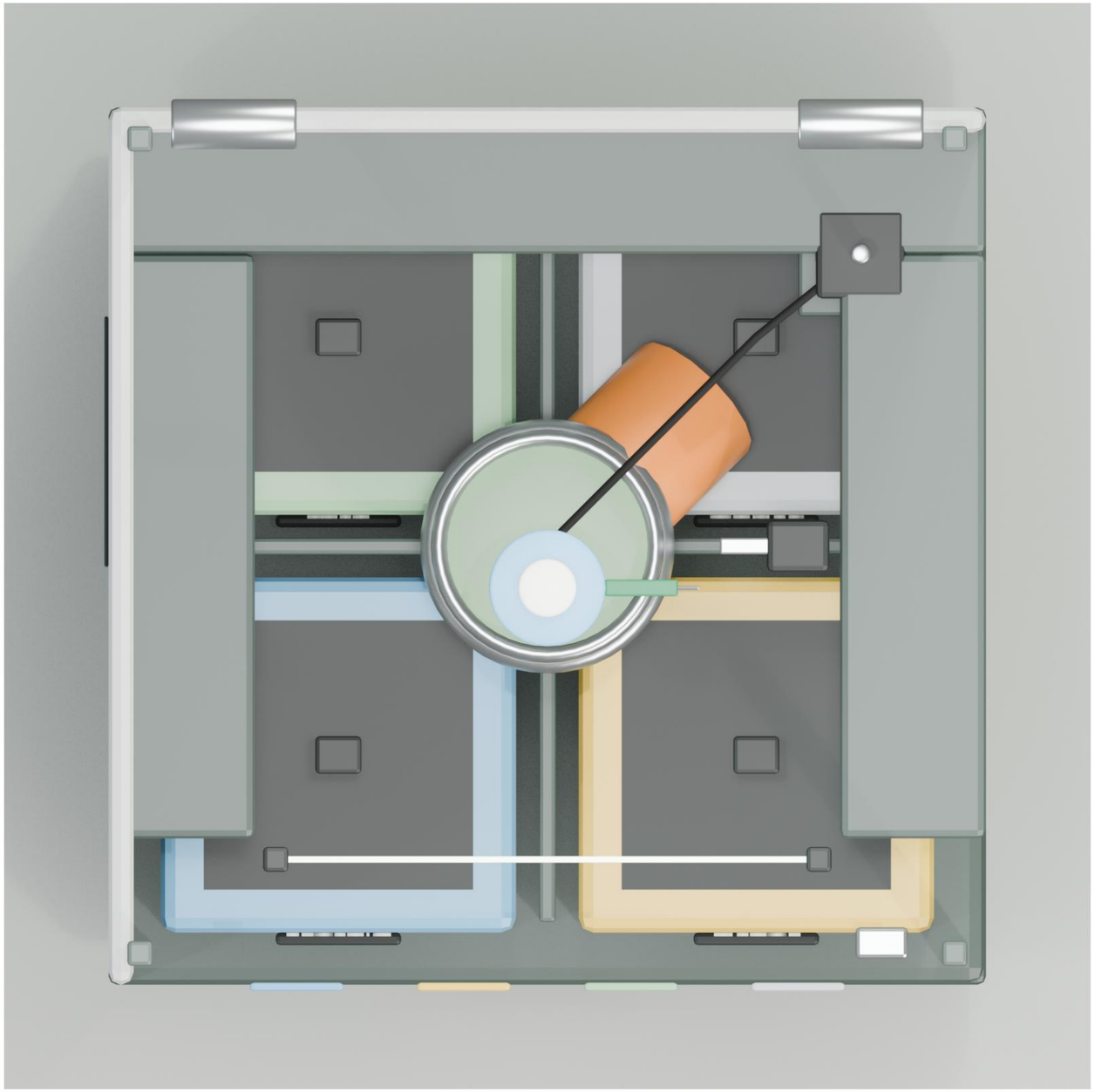
INTERNAL STRUCTURE
內部結構

1. Fixed weigh platform
   固定稱重平台
2. 5 kg load cell
   負重感測器
3. MG996R servo
   伺服馬達
4. Funnel 220→75 mm
   漏斗
5. NEMA 17 + 3:1 belt
   步進馬達 + 皮帶
6. 75 mm rotating spout
   旋轉滑槽
7. 4 bins 170×170×300
   四個桶
8. IR safety beam / bin IR
   紅外對射 / 滿度
9. ESP32-S3 / HX711 / A4988
   電子倉

Concept model / 概念模型

TOP VIEW / BIN LAYOUT

2 × 2 bins, no corner notches; spout rotates above bin tops

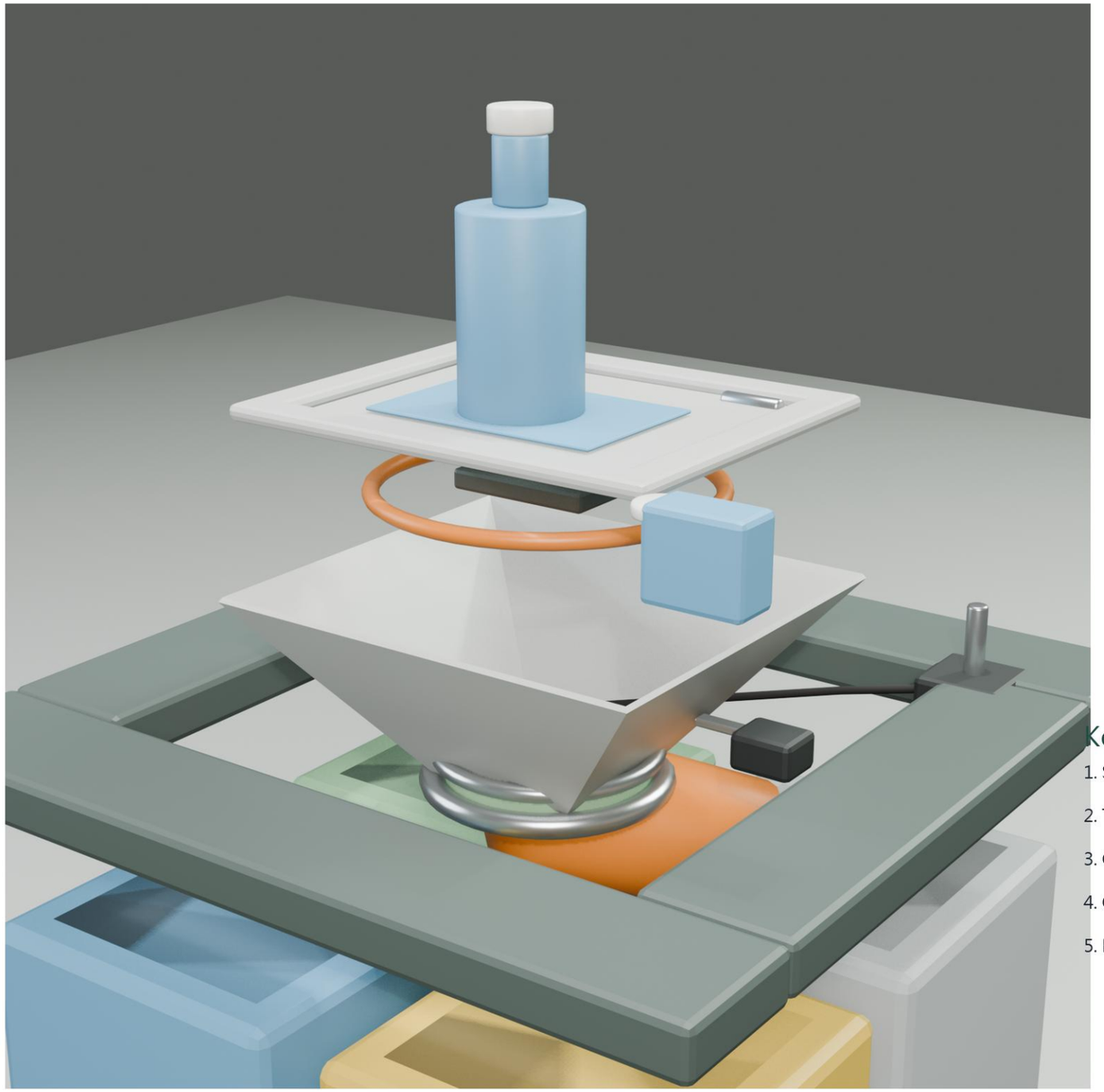


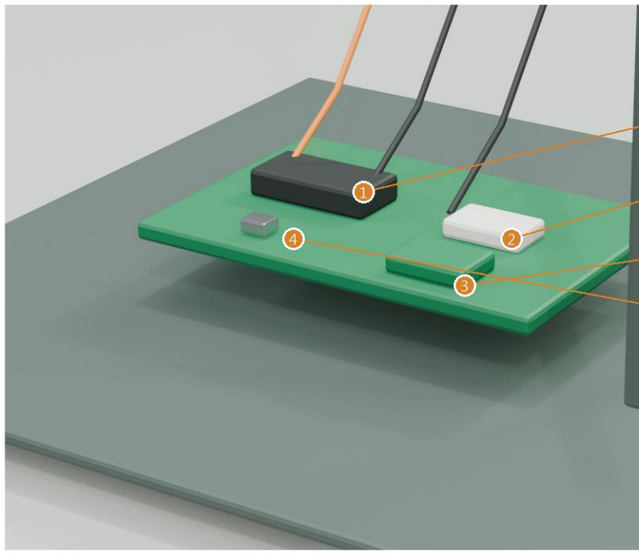
Bin layout

Bin	Centre (mm)	Colour
1 Plastic	(-100, -100)	Blue
2 Metal	(+100, -100)	Yellow
3 Wet	(-100, +100)	Green
4 Other	(+100, +100)	Grey

Sensors

Device	Location
ESP32-S3-CAM	Lid underside
IR beam	Hopper mouth
Load cell	Under platform
Moisture pad	On trapdoor
4 × IR full	Mid-frame / bin
Home switch	Beside spout
OLED / SD	Front / camera





Key checks

1. Spout torque: 1.35 N·m available vs ~0.34 N·m required.
2. Trapdoor: MG996R + spring; item <=0.5 kg.
3. Clearance: 225 mm tray-to-lid; upright 500 mL bottle.
4. On-device model: MobileNetV2 0.35/0.5, INT8.
5. Build frame/bins first; fit platform/door before lid.